DIC09
NAND and NOR Logic Gates

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 9 (PDF pages 35–38): size a reference inverter, then compare two-input NAND/NOR input order, dimensions, transfer curves, noise margins, speed and power.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: balance inverter (35,38)	01_inverter_sizing	Ratios 3–7; select 6
Steps 2/3: NAND/NOR sizing and DC (36–38)	02_nand, 03_nor, 04_sizing_refinement	A, B, A=B; complete NM
Questions 1/2: late input and parasitics (38)	One input switches after the other settles	Both input positions measured
Question 3: factors 1.5/2/2.5/3 (38)	Four cases in 02/03	Compare output rise/fall
Question 4: speed, power and area (38)	Average supply power and total widths	Equal loads; area is only a proxy

3 Circuit and design derivation

Length is fixed at $0.25\,\mu$m and reference $W_n = 0.6\,\mu$m. Sweep $W_p/W_n = 3, 4, 5, 6, 7$ and choose 6 as the gate reference. NAND has series NMOS and parallel PMOS; NOR reverses these networks. A is nearest the output in the NAND stack and nearest the supply in the NOR stack. Each gate retains simultaneous A=B switching and the two individually sensitized inputs. The inactive input is held at 1 for NAND and 0 for NOR.

Initially, series widths are multiplied by K=1.5/2/2.5/3 to test $R \propto L/W$. Input-specific refinement then compensates for parasitic effects. The selected third refinement uses NAND PMOS A/B widths 4.14/5.04 and NMOS widths 1.5 each; NOR PMOS widths are 21.6 each and NMOS A/B widths

2.7/1.02, in μm. The source Lab9 first file incorrectly referenced an external PWM circuit; a self-contained inverter calibration replaces it. All gates include their own circuitry and measurements.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Every output has 20 fF. Input edges are 0.1 ns, period 5 ns; simulate 15 ns with 1 ps printing. DC is 0–1 V in 0.5 mV steps. Supply power is averaged over full cycles from 5–15 ns and excludes ideal input-source energy. NM uses the Lab8 boundary-output definition. The handout specifies approximate equality without a tolerance; 10% is an added engineering comparison target, not a claimed original acceptance threshold.

5 Results and specification checks

All four decks and 16 cases completed without failed measurements.

W_p/W_n	V_M (V)	t_r (ps)	t_f (ps)
3	0.4825	222.64	110.85
4	0.4956	173.83	113.49
5	0.5058	144.13	115.26
6	0.5142	124.85	117.32
7	0.5212	110.93	119.71

Gate	Input	V_M (V)	NM_L (V)	NM_H (V)
NAND	both	0.5464	0.4615	0.3621
NAND	a	0.4923	0.3924	0.4226
NAND	b	0.4827	0.3808	0.4331
NOR	both	0.4792	0.3849	0.4430
NOR	a	0.5455	0.4539	0.3585
NOR	b	0.5352	0.4442	0.3712

Selected design	t_r (ps)	t_f (ps)	t_{pHL} (ps)	t_{pLH} (ps)
INV	123.30	116.03	109.74	96.30
NAND A	130.26	111.14	99.02	102.50
NAND B	124.24	111.18	107.10	104.32
NOR A	134.89	122.00	135.99	121.49
NOR B	128.66	116.35	150.46	79.97

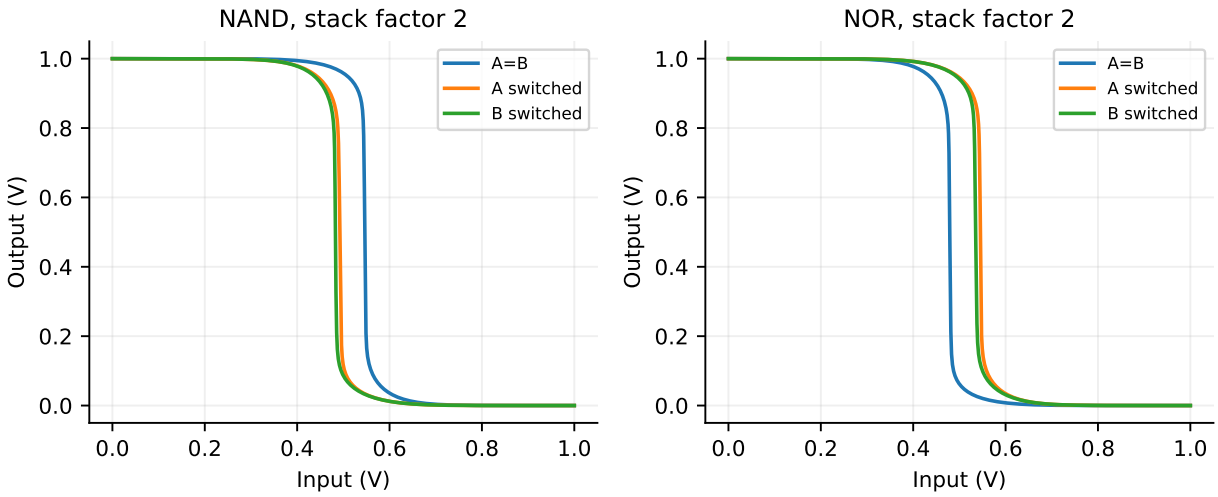


Figure 1: Three input conditions with stack factor 2; redrawn from this run's HSPICE data.

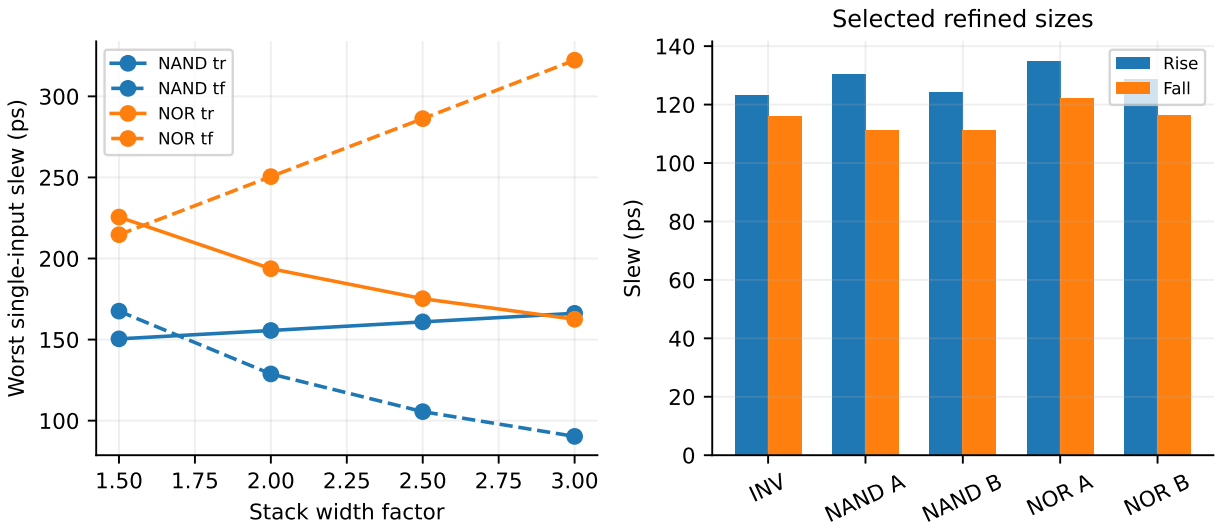


Figure 2: Stack-factor sweep and final size refinement; redrawn from this run's HSPICE data.

Mean supply power and total widths at initial K=2:

Gate	$P_{A=B}$ (μW)	P_A (μW)	P_B (μW)	$\sum W$ (μm)
NAND	7.145	6.480	7.299	9.6
NOR	9.134	12.916	7.232	15.6

6 Discussion

The ratio-6 inverter has a threshold near 0.514 V and rise/fall times of about 124.85/117.32 ps: close, not perfectly symmetric. In the selected refinement, the maximum single-input slew deviation from the same-deck inverter is about 9.4%. Simultaneous activation of parallel devices is faster, so equality across every input mode is not expected.

The latest signal should drive the series device nearest the output: A for NAND, B for NOR. The other input is held settled in this test. Internal-node charging and body effect make simple stack-count width multiplication inexact; widening also increases junction, overlap and input capacitance. At initial K=2, total widths are 9.6 for NAND and 15.6 μm for NOR. This is an active-width proxy at equal L, excluding layout, contacts and routing. CSV files retain average supply power for every input mode. NOR is generally slower and more energy-consuming in this equal-load comparison, but no universal cross-process ranking is claimed. Matching slew by greatly enlarging NOR trades area and input loading for speed.

7 Conclusion

All three steps and four questions have corresponding tests. The specified multipliers are retained before further refinement. Approximate 10% matching is a result against this run's added engineering target; original-process values and unknown reliability constraints are not declared satisfied.

8 MOS transistor schematics

Source: `02_nand.sp` and `03_nor.sp`. These are the complete four-MOS GATE cells, with their output load. Separate instances test both inputs switching or one fixed input. $L = 0.25\,\mu$m throughout. NAND: $W_p = \mathrm{WPREF}$ and $W_n = 0.6K_{stack}\,\mu$m. NOR: $W_p = \mathrm{WPREF}K_{stack}$ and $W_n = 0.6\,\mu$m. The reference inverter uses the usual two-MOS complementary cell. The sizing-refinement deck retains the four-MOS NAND/NOR topologies shown here and independently changes their widths.

The drawings use the device and node names of the cited netlists. Equal net labels denote an electrical connection. Unless stated otherwise, NMOS bulk terminals connect to ground and PMOS bulk terminals connect to V_{DD}. The bulk wires are omitted from the three-terminal symbols for readability.

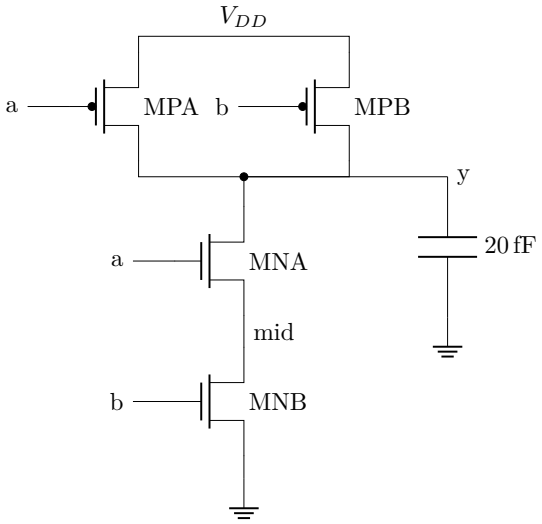


Figure 3: NAND2: parallel PMOS pull-up and series NMOS pull-down. Equal a/b labels connect the corresponding gates.

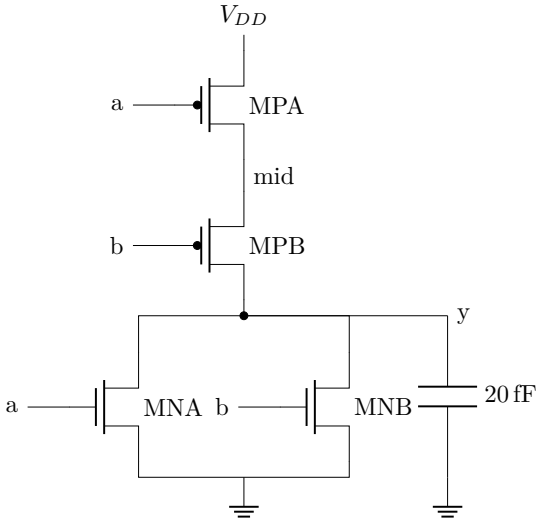


Figure 4: NOR2: series PMOS pull-up and parallel NMOS pull-down. The three test cases set $(a, b) = (in, in), (in, 0), (0, in)$.